

EDUCATION

Ph.D. in Civil and Environmental Engineering , <i>Transport System and Logistics Lab, Imperial College London</i>	Expected 2026
Research Focus: Autonomous Vehicles, Computer Vision, Large Language Models, Reinforcement Learning	
Master of Science in Computer Engineering , <i>New York University</i>	October 2019 - June 2021
Relevant Coursework: Advanced Machine Learning, Algorithmic Machine Learning and Data Science, Information Theory	
Bachelor of Engineering in Computer Science , <i>Shandong University</i>	September 2015 - June 2019
Relevant Coursework: Advanced Programming Language, Pattern Recognition	
Undergrad Summer Visitor , <i>Stanford University</i>	June 2017 - August 2017

RESEARCH EXPERIENCE

Bézier Everywhere All at Once: Learning Drivable Lanes as Bézier Graphs	October 2023
• <i>Hugh Blayney, Hanlin Tian, Hamish Scott, Nils Goldbeck, Chess Stetson, Panagiotis Angeloudis</i> • Accepted to CVPR 2024. • Developed a method for large-scale extraction of lane topology from aerial imagery, utilizing Bézier curve parameterization and a transformer-based model. • Achieved competitive results on the Urban Lane Graph dataset, producing realistic lane graphs with minimal input, and shared the code for public use.	
Enhancing Autonomous Vehicle Training with Language Model Integration and Critical Scenario Generation	June 2024
• <i>Hanlin Tian, Kethan Reddy, Yuxiang Feng, Mohammed Quddus, Yiannis Demiris, Panagiotis Angeloudis</i> • Submission to IROS 2024. • Introduced CRITICAL, a novel framework for AV training and testing, generating scenarios to address learning gaps in RL agents. • Demonstrated enhanced performance and safety in AV systems, as evidenced in PPO and HighwayEnv simulations.	
Impact of Building-Induced Visibility Restrictions on Intersection Accidents	May 2023
• <i>Hanlin Tian, Yuxiang Feng, Wei Zhou, Anupriya, Mohammed Quddus, Yiannis Demiris, Panagiotis Angeloudis</i> • Presented at the TRB Annual Meeting, TRBAM-24-02409. • Developed a framework for evaluating urban junction safety using OSM and remote sensing data. • Utilized a Graph Neural Network for comprehensive risk assessment, with real-world case studies validating the methodology.	
Vehicle-to-Vehicle Collaborative Navigation for Autonomous Vehicles in Occluded Environments	October 2022
• <i>Leandro Parada, Hanlin Tian, Jose Escribano, Panagiotis Angeloudis</i> • Published in ITSC 2023, DOI: 10.1109/ITSC57777.2023.10422217. • Developed a V2V collaborative control methodology using Proximal Policy Optimisation for enhanced safety in occluded environments. • Validated the approach in a CARLA-based simulation, showing improved performance over traditional strategies.	
Safety Standards for Autonomous Vehicles: Challenges and Way Forward	May 2023
• <i>Elias Nassif, Hanlin Tian, Eduardo Candela, Yuxiang Feng, Panagiotis Angeloudis, Washington Y Ochieng</i> • Presented at ITSC 2023, DOI: 10.1109/ITSC57777.2023.10422153. • Analyzed existing safety standards and proposed a balanced approach for AV standardization. • Developed principles and a strategic plan for promoting standards adoption in the AV industry.	
The Robust Semantic Segmentation UNCV2023 Challenge	October 2023
• <i>Hanlin Tian, Kenta Matsui, Tianhao Wang, Fahmy Adan</i> • Presented in ICCV Workshops 2023, article on pp. 4618-4628. • Secured 6th place in the UNCV2023 Challenge. • Implemented a snow simulation data augmentation technique for improved segmentation in winter conditions. • Integrated the Xception architecture into our segmentation model for enhanced performance.	

Active Crowd Analysis for Pandemic Risk Mitigation for Visually Impaired Individuals

July 2020

- Samridha Shrestha, Daohan Lu, **Hanlin Tian**, Qiming Cao, Julie Liu, John-Ross Rizzo, William H. Seiple, Maurizio Porfiri, Yi Fang
- Presented at ECCV Workshops 2020, article on pp. 422-439.
- Developed a smartphone-based neural network for crowd density detection, aiding visually impaired individuals with social distancing.
- Utilized real-time crowd analysis for effective pandemic risk mitigation.

Meta-DCP: Meta-Learning and Deep Learning for 3D Point Cloud Registration

May 2021

- Master's Thesis research.
- Employed Meta-Learning and Deep-Learning techniques for 3D point cloud registration.
- Demonstrated lower loss on unseen point cloud categories compared to traditional methods.

ACTIVITIES

Research Assistant, Imperial College London

September 2023 -

- Engaged in the DeepSafe initiative, focusing on enhancing synthetic data for training AVs in edge-case scenarios to improve performance and reliability.

Research Intern, dRisk

June 2023 - September 2023

- Developed advanced lane graph generation techniques using transformer architectures and Bezier curves for enhanced accuracy and system performance.

Research Assistant, Zhejiang University

October 2022 - January 2023

- Worked with Professor Simon Hu at the Transport Systems & Environment Lab on assembling and evaluating autonomous vehicle models, focusing on hardware assembly and performance analysis.

Research Assistant, New York University

August 2020 - July 2021

- Contributed to the publication "Active Crowd Analysis for Pandemic Risk Mitigation for Visually Impaired Persons" and the development of the "Virtual Touch" assistive platform for visually impaired individuals.
- Authored a Master's thesis, "Meta-DCP," with potential for conference submission.

Software Development Engineer, Chinese Academy of Science

February 2019 - July 2019

- Developed a comprehensive Industrial Vision System and designed a neural network for industrial part inspection using TensorFlow.

Summer Research Assistant, Washington University in St. Louis

July 2018 - September 2018

- Created an Autoencoder for image compression and denoising, achieving promising results in both applications.

EXPERTISE AND SKILLS

Tools and Languages

Research Experiences

Communication

Python, C++, Git, $\text{\LaTeX}$, PyTorch, TensorFlow, R, MongoDB

Machine Learning, Computer Vision, Meta-Learning, Reinforcement Learning

English, Cantonese, Chinese